ShopForMe

Material requirements and specifications

Version 1.3

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1. Objectif

The purpose of this document is to provide an exhaustive definition of the hardware requirements and specifications necessary for the ShopForMe project.

It serves as a reference for the identification, selection, and acquisition of the appropriate hardware, thus ensuring that all hardware components are perfectly aligned with the functional and non-functional objectives of the project.

This document is intended to facilitate understanding and communication between project team members and stakeholders, by providing a clear view of hardware requirements.

It will also help to anticipate the technical and budgetary constraints associated with the hardware aspect of the project.

2. Project Overview

2.1. Project Description

The ShopForMe project aims to transform the shopping experience for blind and partially sighted people through the development of an intelligent supermarket trolley.
Using technologies such as the IoT (Internet of Things), image recognition and assisted navigation, this shopping trolley will be able to guide users through supermarket aisles, detect and inform them of the desired products, and avoid obstacles, making the act of shopping more accessible and autonomous for people with visual impairments.

2.2. Equipment importance in the project

The success of the ShopForMe project depends on a rigorous selection of hardware, as hardware components play a crucial role in the reliability, performance, and user experience of the intelligent shopping trolley.
The accuracy of the sensors, the robustness of the navigation system and the ergonomics of the user interface are directly linked to the quality of the hardware chosen.
So, the realization of our vision depends on our ability to harmoniously integrate advanced technologies into a device that is practical, durable, and easy to use for our target audience.

3. Material requirements

3.1. Identatification of material requirements

This section lists all the hardware components required for the project.

- Trolley: To load the user's purchases as well as housing technological components
- Ultrasound sensors: For obstacle detection and aisle navigation.
- Indoor GPS module: For positioning and navigation inside the shop.
- Camera: For product recognition and label reading.
- Braille touch screen: User interface to enable blind users to navigate the shopping trolley options.
- Microphone and loudspeaker: For voice commands and audio feedback.
- Wi-Fi/Bluetooth module: For connectivity with the mobile application and shop system.
- Rechargeable Battery: To power the shopping trolley during use.
- Raspberry Pi or equivalent: As the main controller to manage input/output and processing.
- Bluetooth Low Energy (BLE) system: localization of the shopping cart.

3.2. Requirements Justification

This section explains the role of each component and its contribution to the project.

- Trolley: Acts as a chassis for the various hardware components of the system.
- Ultrasound sensors: These sensors enable the trolley to detect and avoid obstacles, ensuring safe navigation through the shop's cluttered aisles.
- Indoor GPS Module: Used to precisely guide the shopping trolley through the shop to specific products, using the shop's interior mapping.
- Camera: Allows visual recognition of products and reading of barcodes or labels, facilitating independent shopping.
- Braille touch screen: Provides an accessible interface for blind users so they can interact independently with the shopping trolley, choose products and navigate menus.
- Microphone and loudspeaker: Allows users to control the shopping trolley by voice and receive auditory information about products and navigation.
- Wi-Fi/Bluetooth module: Ensures connectivity between the shopping trolley, the user's mobile application and the shop's IT system to synchronise shopping lists and locate products.
- Rechargeable Battery : Provides sufficient autonomy for the shopping trolley to cover the needs of a complete shopping session without the need for intermediate recharging.
- Raspberry Pi or equivalent: acts as the caddy's brain, controlling all the electronic components and running the software for navigation, product recognition and communication.
- Bluetooth Low Energy (BLE) system:

4. Technical specifications

Component details :
- Name/Type of Component : Server, Sensor, Computer, etc.
- Function: Description of the component's function in the project.
- Technical Characteristics: Detailed specifications (capacity, performance, dimensions, etc.).
- Compatibility: Information on compatibility with other components or systems.

4.1. Bluetooth Low Energy (BLE) system

4.1.1. BLE Location System Operation

- **BLE Beacons :**
  - Function: Emits BLE signals for location inside the shop.
  - Location: Strategically placed in the shop to cover all important areas.
  - Signal: Constantly emit signals that are picked up by a receiver.

- **BLE receiver :**
  - Function: Receives BLE signals from beacons to detect position.
  - Placement: Installed on the trolley or used via a smartphone.
  - Method: Uses the intensity of the signal received (RSSI) to estimate the position.

- **Position calculation:**
  - Function: Estimating the position of the shopping trolley in the shop.
  - Technique: Use of several RSSI tags and application of techniques such as triangulation or fingerprinting.

4.1.2. Technical specifications

- **Utility :**

Precise location of the shopping trolley in the shop, customer guidance, tracking of shopping trolleys, location-based promotions.

- **Accuracy**

Varies from a few decimetres to a few metres, influenced by the density of the beacons, the environment, and the position calculation method.

- **System composition**

BLE beacons:
Number: Varies according to the size of the shop and the desired accuracy.
Unit cost: From a few dollars to around twenty dollars depending on functionality.

BLE receivers:
Types: Customer smartphones (with application) or devices on shopping trolleys.

Server/Management software:
Function: Collection and processing of location data, interaction management.

- **Cost**

Infrastructure: Development of the mobile application/embedded system and data management server.
Maintenance: Replacement of beacons (battery) and software updates.

- **Operating distance**

Range: Up to 50 metres or more, with 10 to 20 metres being optimal for indoor location.

5. System diagram

System Overview: A diagram illustrating how hardware components interact within the project.

Component Integration: An explanation of the integration and interaction between different hardware components.

6. Contraints and Considerations

Material constraints: Specific limitations or challenges related to the material (e.g. environment, space, power supply).

Environmental and Sustainability Considerations: Environmental or sustainability aspects related to the choice of hardware.

6.1. Material Contraints

Type of shop :
Description: The solution is designed primarily for supermarkets and hypermarkets with wide aisles and sufficient space to manoeuvre the shopping trolley.
Limitation: Small shops or shops with narrow aisles may not be compatible due to the size of the shopping trolley and the need for space for navigation.

Environment and Space:
Description: The interior environment of the shop must allow good reception of BLE beacon signals and navigation free of unexpected obstacles.
Limitation: Densely populated or congested areas may interfere with cart navigation and location accuracy.

Power supply:
Description: The shopping cart requires a rechargeable battery with sufficient autonomy to cover the average shopping time in a supermarket.
Limitation: The need to recharge caddies may limit their continuous availability during peak hours.

6.2. Environmental and sustainability considerations

Choice of materials :
Description: Use of recyclable or low environmental impact materials for the manufacture of the shopping trolley and electronic components.
Objective: To minimise the ecological footprint of the product throughout its life cycle.

Energy consumption:
Description: Optimisation of the energy consumption of the BLE system and electronic components to extend battery life.
Objective: To reduce overall energy consumption and encourage efficient use of resources.

Recycling and End of Life:
Description: Planning for the end-of-life of shopping trolleys and electronic components, including the recycling and recovery of materials.
Objective: To facilitate disassembly and recycling, thereby reducing the environmental impact of withdrawal from service.

Adaptability and Updates:
Description: Design of the system to allow software and hardware updates to extend the life of the shopping trolley.
Objective: To avoid early obsolescence and promote a sustainable approach through updates rather than replacements.

7. References and normes

Applicable standards: List the industrial or regulatory standards relevant to the equipment.

Technical documentation : References to technical documents, manuals or component datasheets.

8. Annexes

Component data sheets: Include manufacturers' data sheets or brochures for key components.

Glossary: Definitions of technical terms used in the document.